

Lab 1 - Build an Enterprise HR Assistant Agent using Microsoft Copilot Studio

Business Scenario

Contoso Enterprises is a multinational organization with over 15,000 employees distributed across multiple countries. The HR department receives hundreds of repetitive employee requests every day regarding leave policies, travel reimbursement, employee benefits, onboarding, work-from-home guidelines, payroll, and employee profile information.

The organization wants to build an Enterprise HR Assistant using Microsoft Copilot Studio that can:

- Answer HR policy questions using enterprise documents (RAG)
- Retrieve employee information from an external source
- Provide grounded responses
- Trigger business actions
- Escalate unsupported requests
- Be reusable across Microsoft Teams and Web channels

The solution must be vendor-managed through Microsoft Copilot Studio while using enterprise governance and security practices.

Project Goal

After completing this lab, you will be able to:

- Create an AI Agent in Microsoft Copilot Studio
 - Configure Generative AI
 - Build a RAG Knowledge Base
 - Connect external employee data
 - Test enterprise AI responses
 - Understand Agent Orchestration
-

Prerequisites

- Microsoft 365 Account
 - Microsoft Copilot Studio License
 - Power Platform Environment
 - OneDrive for Business
-

Dataset Overview

Knowledge Documents

- Employee Handbook
- HR Policy
- Leave Policy
- Travel Policy

- Benefits Guide
- Code of Conduct

Employee Dataset

EmployeeMaster.xlsx

Step 1: Verify Microsoft Copilot Studio Access

Execution

Open the Microsoft Copilot Studio portal

<https://copilotstudio.microsoft.com/>

Sign in using your Microsoft 365 account.

Verify that the Copilot Studio home page loads successfully.

Confirm that you can access the available environments.

Step 2: Verify Your Power Platform Environment

Execution

Navigate to the Environment selector.

Review the available environments.

Confirm:

- Environment Name
- Region
- Environment Type
- Managed or Default Environment

Record the information.

Explanation

Every Copilot Studio agent is deployed into a Power Platform environment. Understanding the environment configuration is essential before creating enterprise solutions.

Expected Outcome

The active environment is identified and documented.

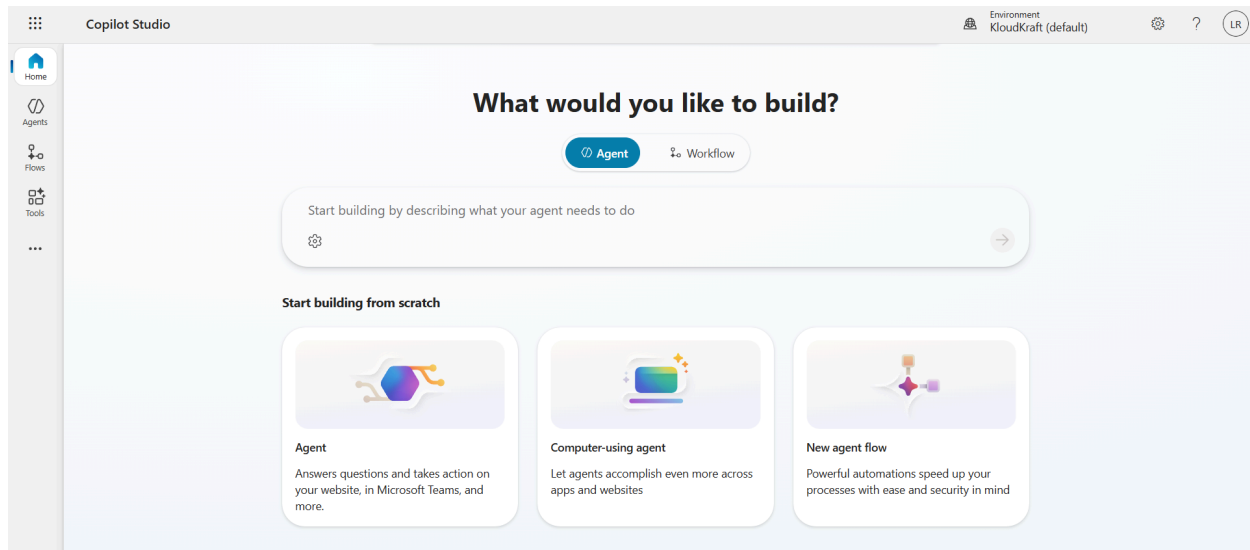
Step 3: Validate Available AI Features

Execution

Navigate through the Copilot Studio interface and verify whether the following options are available:

- Create Agent
- Knowledge
- Topics
- Actions
- Analytics
- Channels

- Publish



Expected Outcome

A list of available Copilot Studio features is documented.

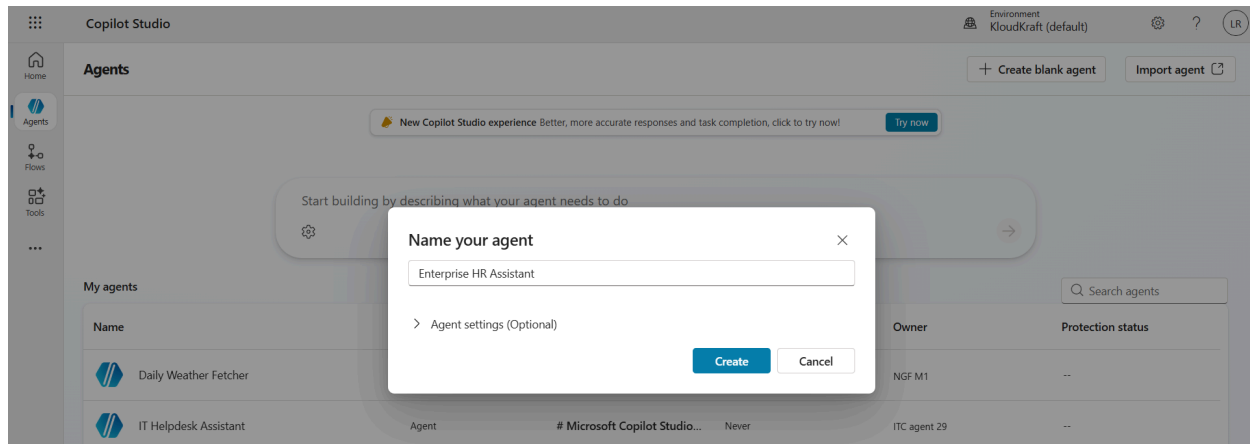
Step 4: Create a New AI Agent

Execution

Select **Create Agent**.

Provide the following information:

- Agent Name: **Enterprise HR Assistant**
- Description: **AI-powered HR support assistant for employees**
- Language: English



Create the agent.

Explanation

The agent serves as the central orchestration component for HR-related conversations and workflows.

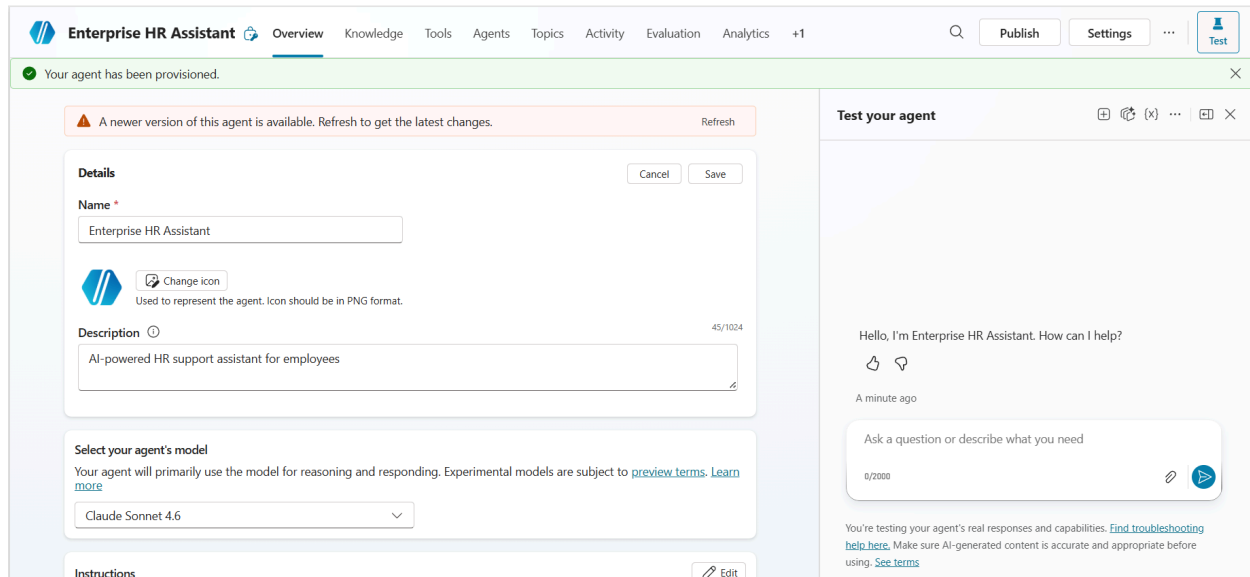
Step 5: Explore the Agent Workspace

Execution

Review the available sections within the newly created agent, including:

- Overview
- Instructions
- Knowledge
- Topics
- Actions
- Analytics

- Channels
- Publish



Explanation

Understanding the workspace helps you navigate and configure the agent efficiently.

Step 6: Configure the Agent Name and Description

Execution

Open the agent settings and verify:

- Agent Name
- Description
- Default Language

Update the description to reflect the HR support use case.

Explanation

A clear description helps administrators understand the purpose of the agent.

Step 7: Configure Agent Instructions

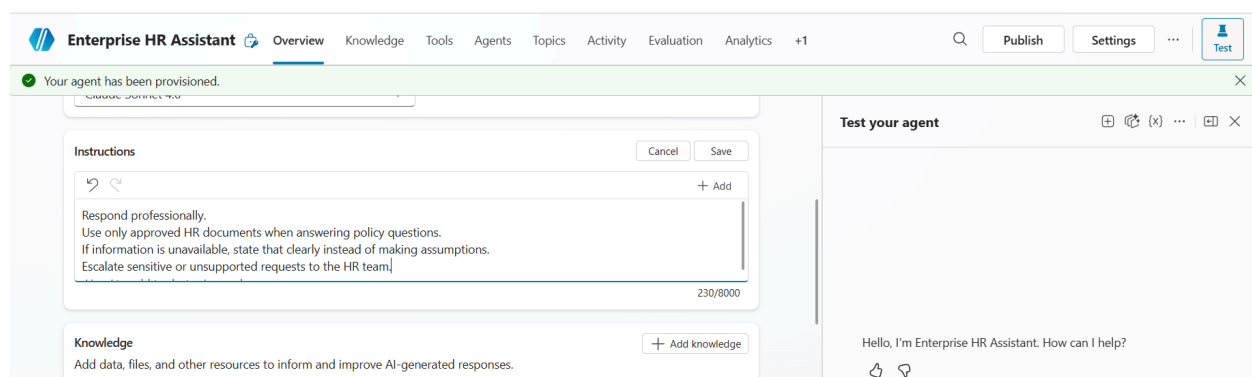
Execution

Open the **Instructions** section and define the agent's behavior.

Include guidance such as:

- Respond professionally.
- Use only approved HR documents when answering policy questions.
- If information is unavailable, state that clearly instead of making assumptions.
- Escalate sensitive or unsupported requests to the HR team.

Save the instructions.



Explanation

Instructions define the overall behavior and response style of the AI agent.

Step 8: Review Knowledge Source Options

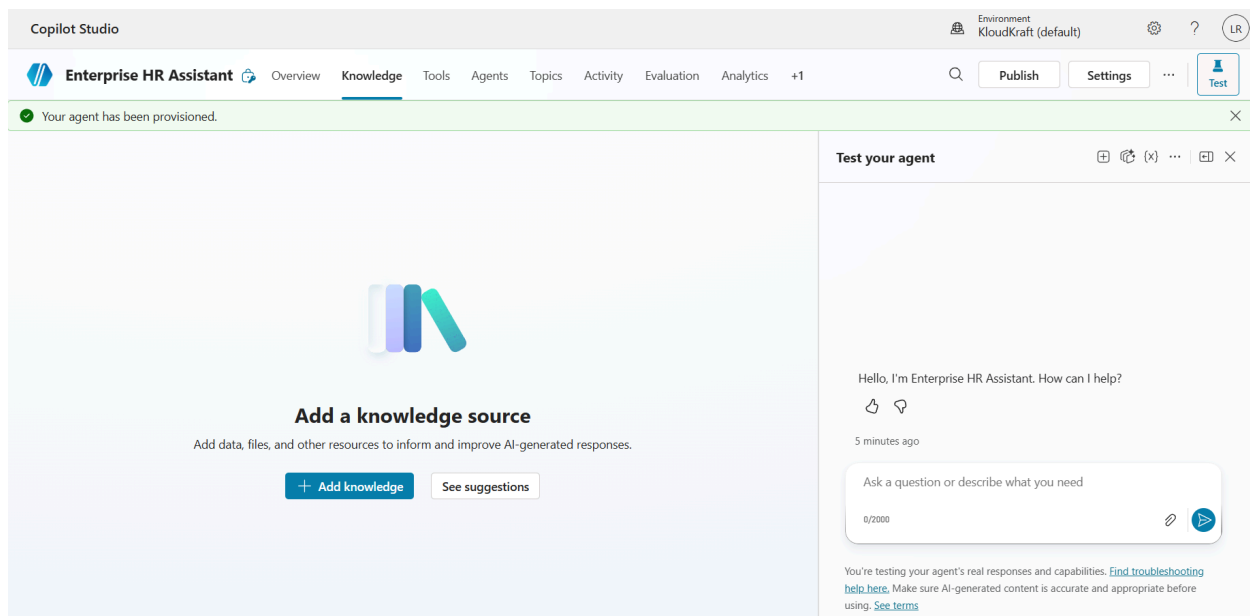
Execution

Navigate to the **Knowledge** section.

Review the supported knowledge source types available in your environment, such as:

- Uploaded Files
- SharePoint
- OneDrive
- Public Websites
- Dataverse (if available)

Document the available options.



Explanation

Knowledge sources provide the foundation for Retrieval-Augmented Generation (RAG).

Expected Outcome

Available knowledge source types are identified.

Step 9: Review the Enterprise Dataset

Execution

Review the lab dataset and identify:

Knowledge Documents

- Employee Handbook.docx
- HR Policy.docx
- Leave Policy.docx
- Travel Policy.docx
- Benefits Guide.docx
- Code of Conduct.docx

Employee Dataset

- EmployeeMaster.xlsx

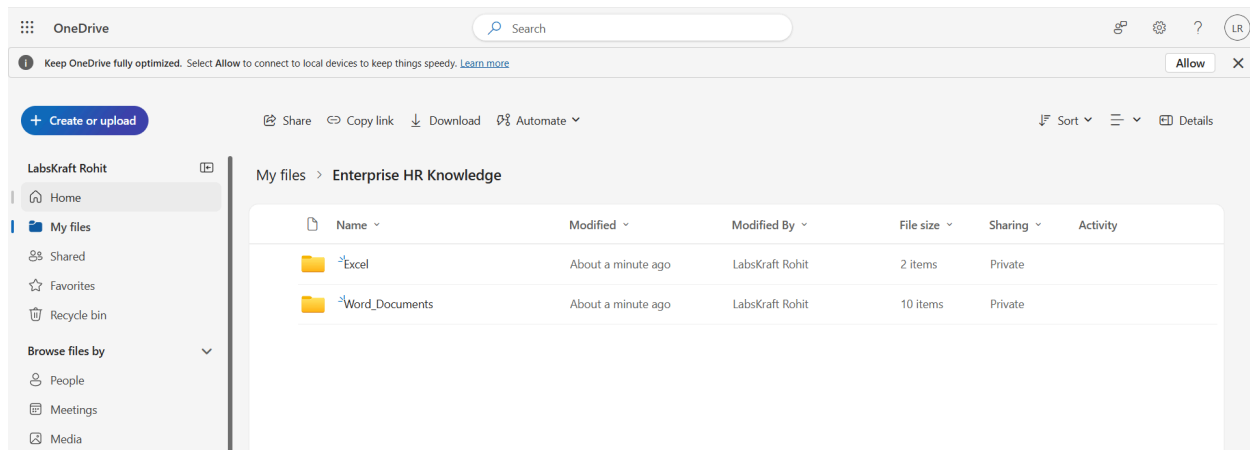
Explanation

Understanding the available datasets before configuration helps you decide which files will be used for knowledge retrieval and which will support business data operations.

Step 10: Create an Enterprise Knowledge Repository

Execution

1. Sign in to your Microsoft 365 account.
2. Open **OneDrive**
3. Create a new folder named **Enterprise HR Knowledge**.
4. Open the folder and keep it ready for uploading the lab datasets.



Explanation

A dedicated repository centralizes enterprise documents, simplifies knowledge management, and makes it easier to replace the storage connector in future implementations.

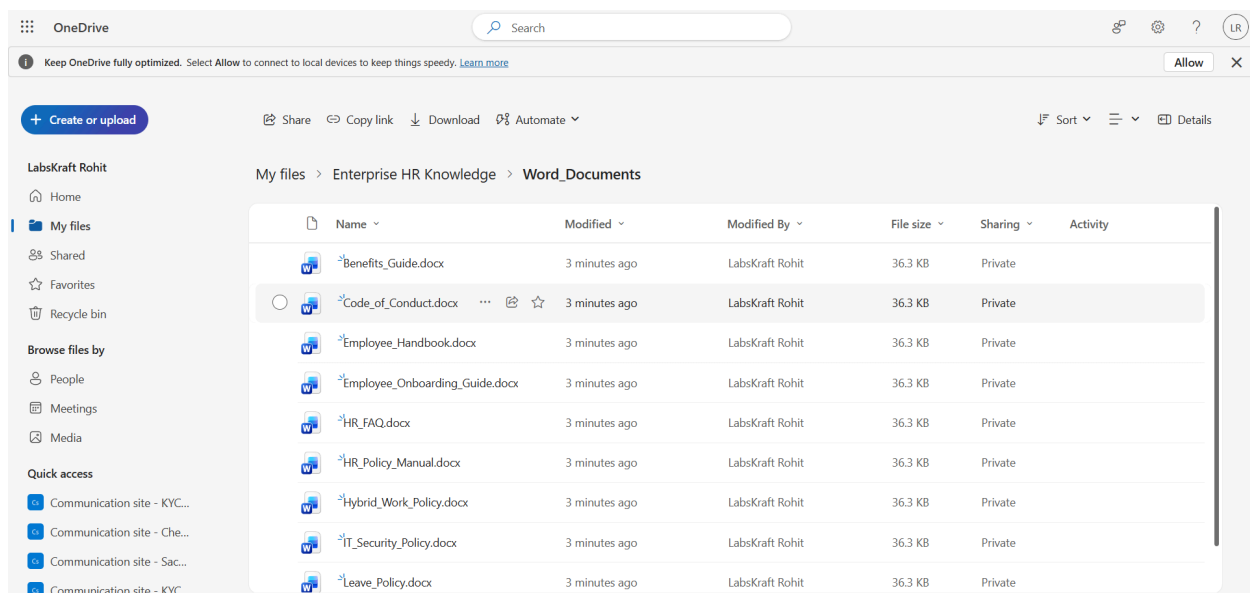
Step 11: Upload HR Knowledge Documents

Execution

Upload the following documents to the **Enterprise HR Knowledge** folder:

- Employee Handbook.docx
- HR Policy.docx
- Leave Policy.docx
- Travel Policy.docx
- Benefits Guide.docx
- Code of Conduct.docx

Wait until all uploads complete successfully.



Explanation

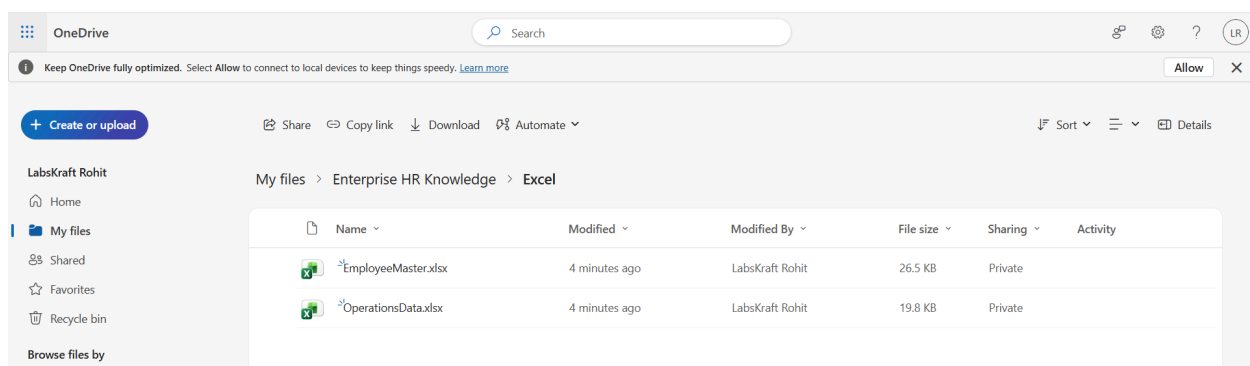
These documents will become the knowledge base used by the agent to answer HR policy questions through Retrieval-Augmented Generation (RAG).

Step 12: Upload the Employee Dataset

Execution

Upload **EmployeeMaster.xlsx** to the same OneDrive folder.

Verify that the file is visible along with the HR documents.



Explanation

The employee dataset will be used later in the lab when configuring business actions and employee information retrieval.

Step 13: Review the Enterprise Documents

Execution

Open each HR document and review its content.

Confirm that the documents include relevant HR information such as:

- Leave policies
- Employee benefits

- Travel reimbursement
- Code of conduct
- Onboarding guidelines
- Workplace policies

Do not modify the content.

Explanation

Reviewing the documents helps you understand the type of information the agent will retrieve during conversations.

Step 14: Review the Employee Dataset

Execution

Open **EmployeeMaster.xlsx** and review the available columns.

Identify key employee attributes such as:

- Employee ID
- Employee Name
- Department
- Designation
- Email Address
- Manager
- Office Location
- Employment Status

Close the workbook after reviewing it.

Explanation

Understanding the dataset structure makes it easier to map employee information when building business actions later in the lab.

Knowledge Source Configuration

Step 15: Open the Knowledge Workspace

Execution

1. Return to **Microsoft Copilot Studio**.
2. Open the **Enterprise HR Assistant** agent.
3. Select **Knowledge** from the left navigation menu.
4. Review the Knowledge workspace.

Explanation

The Knowledge workspace is used to connect enterprise repositories that the agent can search to generate grounded responses.

Step 16: Add a New Knowledge Source

Execution

1. Select **Add Knowledge**.
2. Review the available knowledge source options.
3. Select **OneDrive** as the connector for this lab.

Note: In future implementations, this step can be replaced with SharePoint, Azure Blob Storage, Dataverse, SQL Server, or another supported connector.

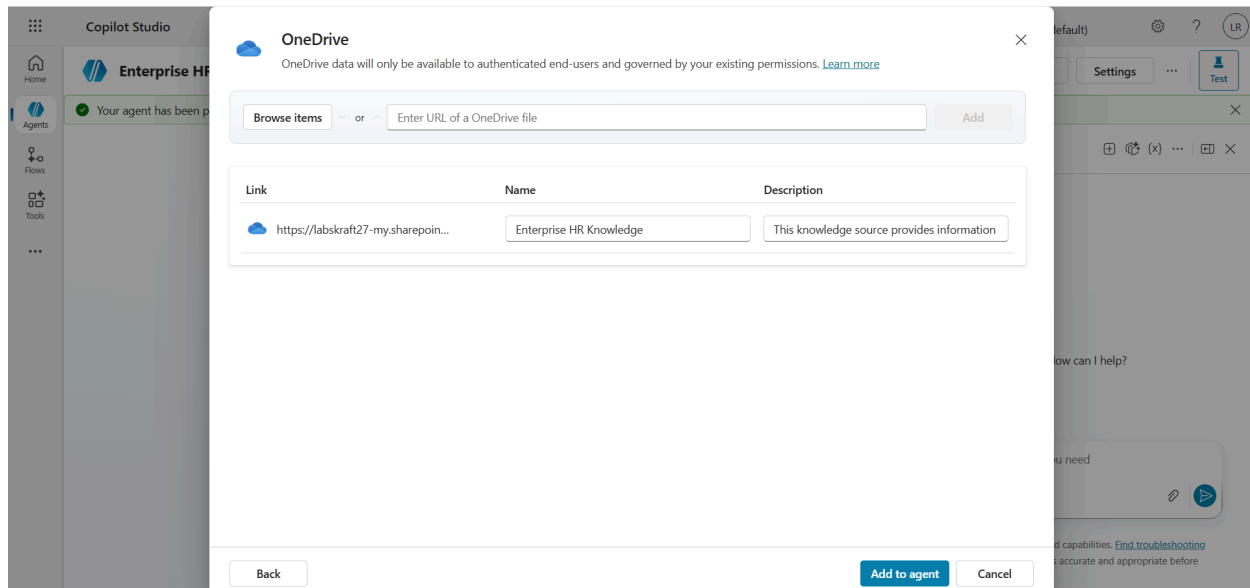
Explanation

Copilot Studio supports multiple enterprise connectors. The connection process remains similar regardless of the selected repository.

Step 17: Select the Enterprise Repository

Execution

1. Browse your OneDrive folders.
2. Select the **Enterprise HR Knowledge** folder.
3. Confirm the folder selection.
4. Continue to the next screen.



Explanation

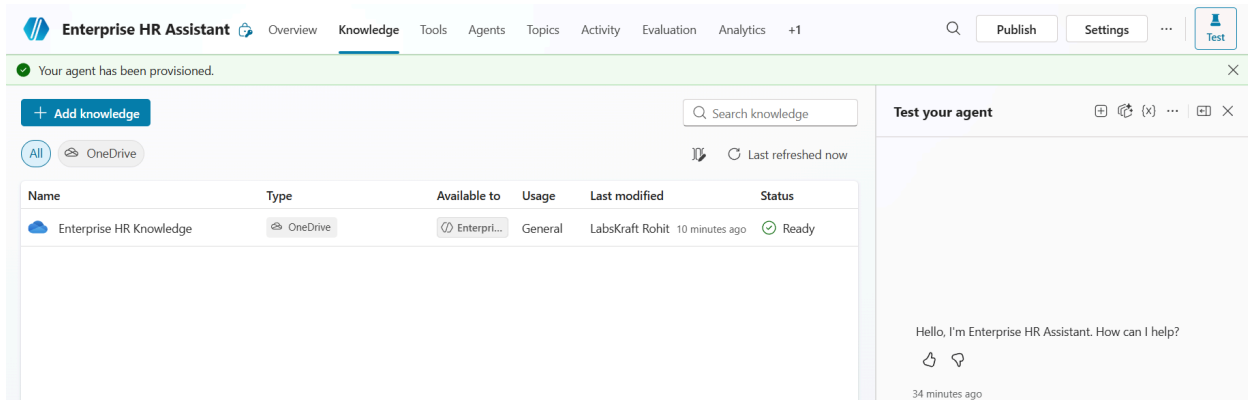
Selecting the folder allows Copilot Studio to index all supported documents stored within that location.

Knowledge Validation

Step 18: Validate the Enterprise Knowledge Base

Execution

1. Open the **Knowledge** section.
2. Verify that the connected OneDrive repository is active.
3. Confirm that synchronization has completed successfully.
4. Review the indexed HR documents.
5. If synchronization is still in progress, wait until it completes before proceeding.



Explanation

This validation ensures the enterprise knowledge repository is connected and all approved HR documents are available for Retrieval-Augmented Generation (RAG).

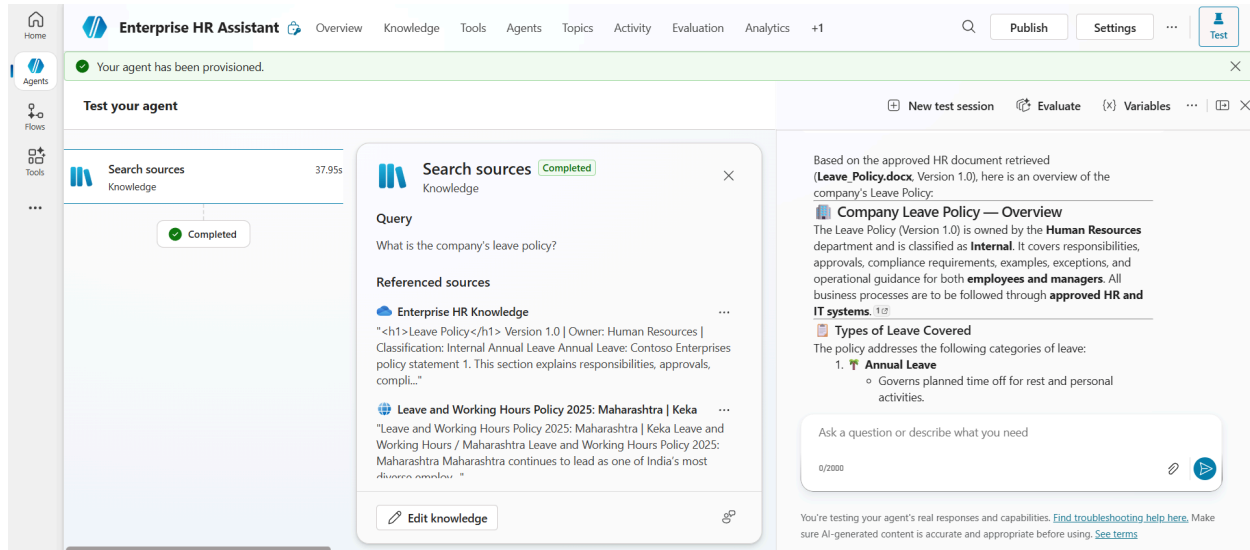
Step 19: Validate Knowledge Retrieval

Execution

Open the **Test your agent** panel and ask questions from different policy categories, such as:

- What is the company's leave policy?
- What employee benefits are available?
- How does travel reimbursement work?
- What are the workplace code of conduct guidelines?

Review the responses and verify they align with the uploaded HR documents.



Explanation

Testing multiple policy categories confirms that the agent retrieves information from the enterprise knowledge base instead of generating unsupported responses.

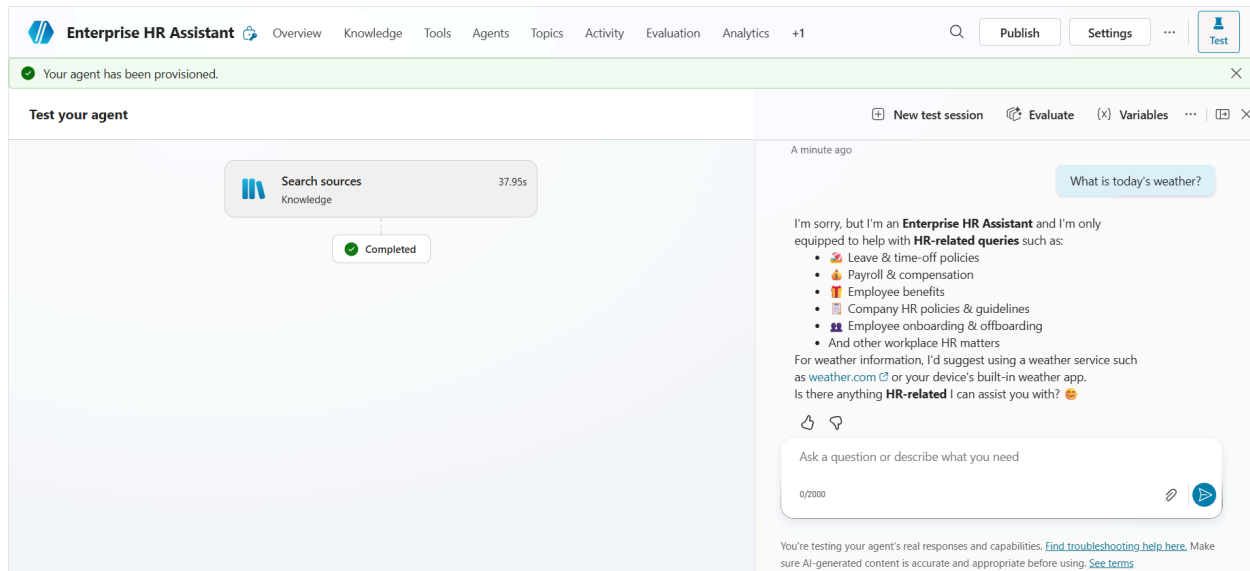
Step 20: Validate Unsupported Requests

Execution

Ask questions that are not covered by the uploaded documents, for example:

- What is today's weather?
- What is the company's stock price?
- Who won the last football world cup?

Review how the agent responds.



Explanation

Enterprise copilots should acknowledge when information is unavailable rather than providing speculative or unrelated answers.

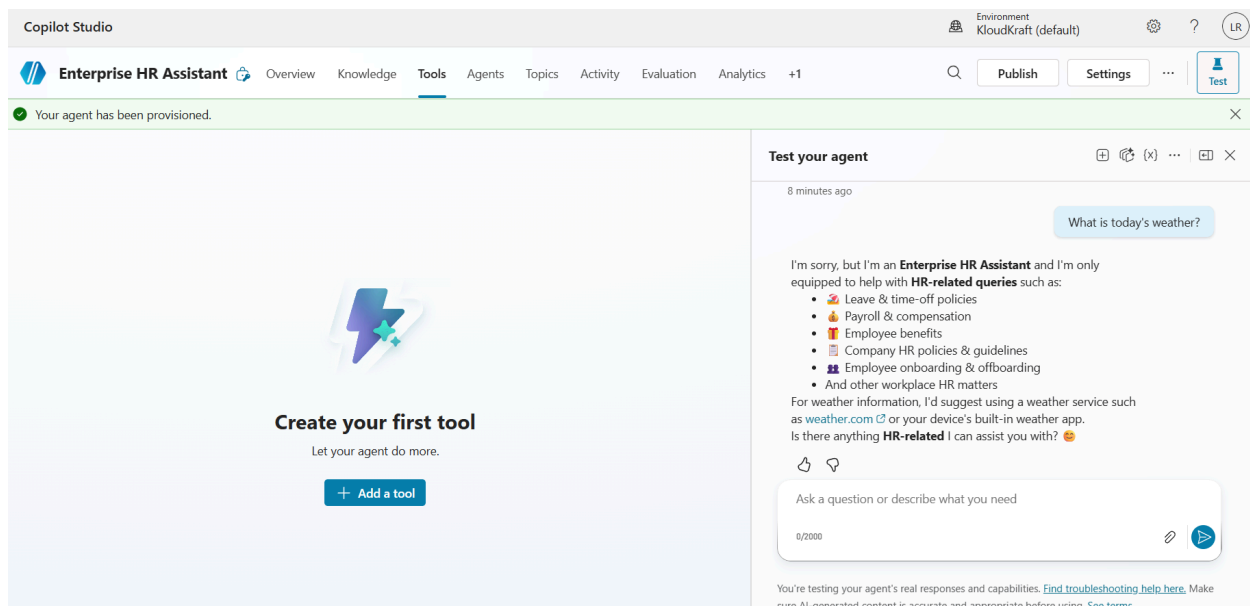
Employee Data Integration

Step 21: Review Available Enterprise Connectors

Execution

1. Open the **Tools** section.
2. Review the available connectors in your environment.
3. Identify connectors that can access enterprise business data, such as:
 - Excel Online (Business)
 - SharePoint
 - Dataverse

- SQL Server
 - Power Automate
4. Note which connectors are available in your tenant.



Explanation

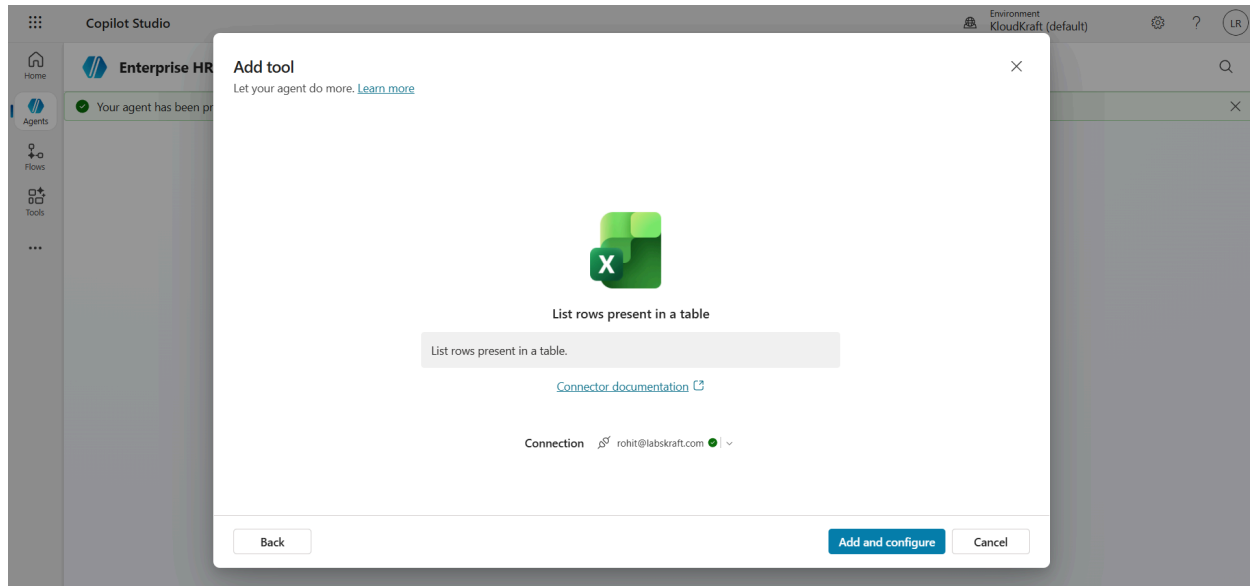
Copilot Studio uses connectors and actions to retrieve and update business data. The available options depend on your licensing and environment configuration.

Step 22: Create an Employee Lookup Tool

Execution

1. Open **Tools**.
2. Select **Add a Tool**.
3. Search for **Excel Online (Business)**.
4. Select **List rows present in a table**.

5. Sign in if prompted.



Explanation

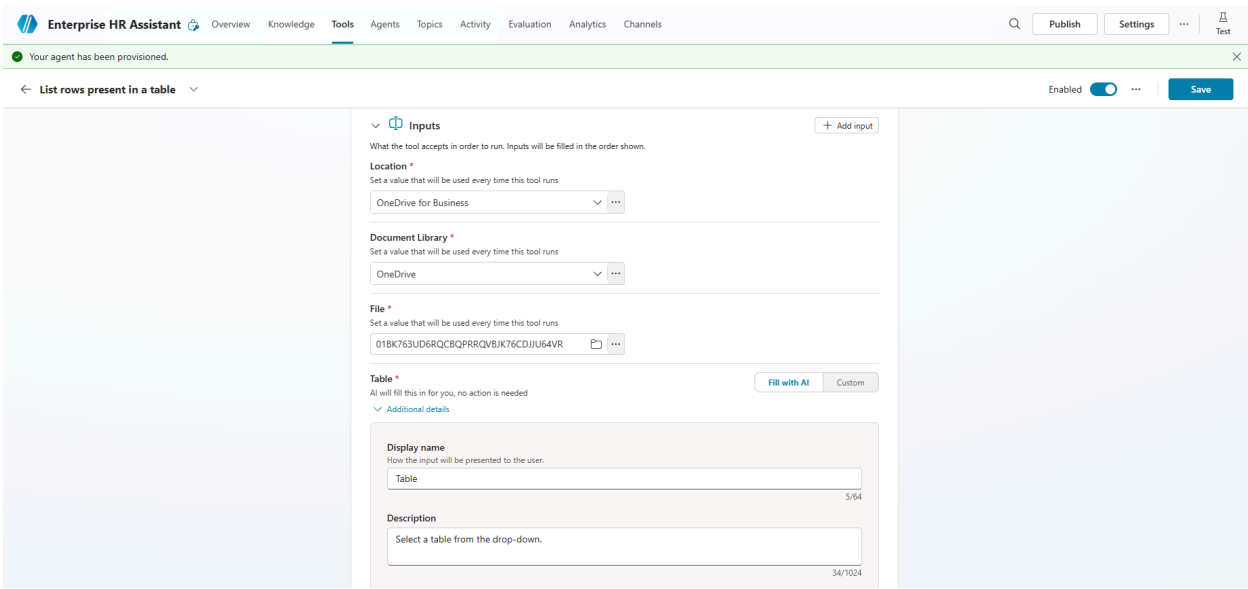
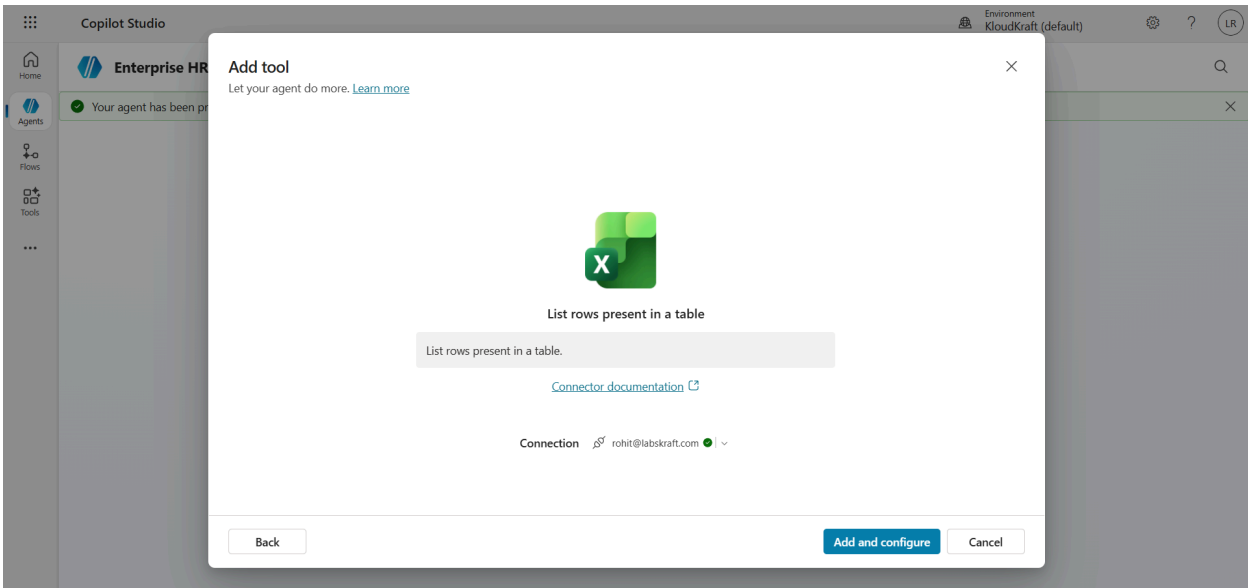
Tools extend the agent by connecting it to external business systems. In this lab, the Excel Online (Business) tool retrieves employee records stored in `EmployeeMaster.xlsx`.

Step 23: Create an Employee Lookup Action

Execution

1. In the **Tools** section, create a new action.
2. Select the connector that will access the employee dataset (Excel Online (Business)) for this lab).
3. Authenticate if prompted.
4. Select **EmployeeMaster.xlsx** as the data source.

5. Save the action.



Explanation

Tools allow the agent to interact with structured enterprise data instead of relying solely on document-based knowledge.

Conversation Design

Step 24: Review Default Topics

Execution

Open the **Topics** section and review the default system topics, including greetings, fallback handling, and conversation endings. Do not modify them.

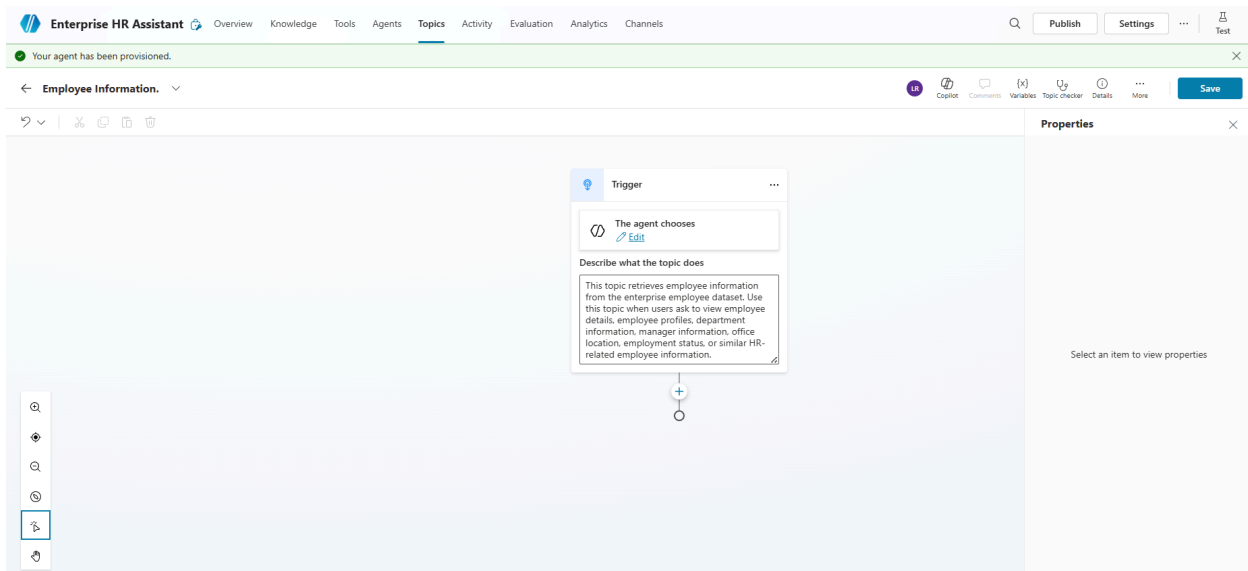
Explanation

System topics provide foundational conversational behavior that can be extended with custom enterprise scenarios.

Step 25: Create an Employee Lookup Topic

Execution

1. Create a new topic named **Employee Lookup**.
2. Add trigger phrases such as:
 - Show my employee details
 - Find employee information
 - View employee profile
3. Save the topic.



Explanation

Custom topics allow the agent to recognize specific business requests and route them to the appropriate action or workflow.

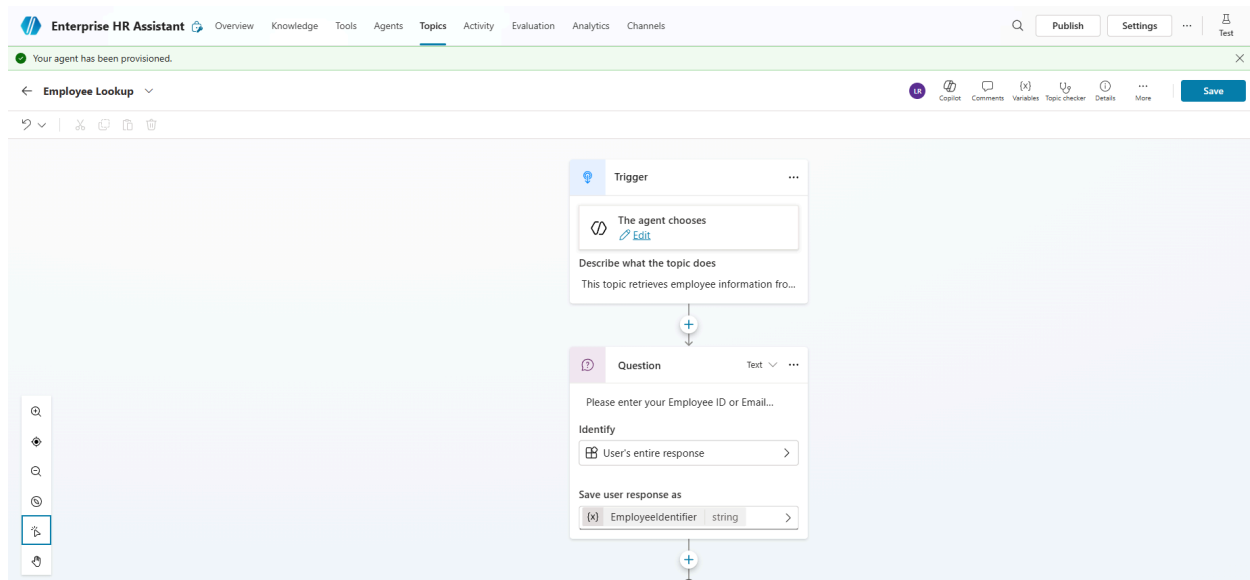
Conversation Design

Step 26: Collect the Employee Identifier

Execution

1. Open the **Employee Lookup** topic.
2. Select the + icon below the **Trigger** node.
3. Choose **Question**.
4. Configure the question using the following values:
 - **Question:** Please enter your Employee ID ?
 - **Response Type:** Text

- **Identify:** User's entire response
 - **Save user response as:** EmployeeIdentifier
 - **Data Type:** String
5. Save the topic.



Explanation

This step captures the Employee ID entered by the user and stores it in the **EmployeeIdentifier** variable. The variable is later used by the Excel connector to retrieve the matching employee record.

Step 27: Add the Excel Employee Lookup Connector

Execution

1. Select the + icon below the **Question** node.
2. Choose **Tool**.
3. Search for **Excel Online (Business)**.
4. Select **Get a row**.

5. Configure the connector as follows:

Property	Value
Location	OneDrive for Business
Document Library	OneDrive
File	EmployeeMaster.xlsx
Table	Table1
Key Column	EmployeeID
Key Value	EmployeeIdentifier (dynamic variable captured in Step 26)

6. Save the topic.

The screenshot displays the 'Enterprise HR Assistant' interface. The main workspace shows a flowchart where the 'Get a row' connector is configured. The connector is set to 'Excel Online (Business)' and is configured to retrieve data from 'EmployeeMaster.xlsx' in 'Table1'. The 'Key Column' is set to 'EmployeeID' and the 'Key Value' is set to 'EmployeeIdentifier'. The 'Location' is 'OneDrive for Business' and the 'Document Library' is 'OneDrive'. The 'File' is 'EmployeeMaster.xlsx'. The 'Table' is 'Table1'. The 'Key Column' is 'EmployeeID' and the 'Key Value' is 'EmployeeIdentifier'. The 'Advanced parameters' section shows 'DateTime Format' set to 'String'.

Explanation

The **Get a row** connector searches the **EmployeeMaster.xlsx** file using the Employee ID provided by the user and retrieves the matching employee record from the Excel table.

Step 28: Test Employee Lookup

Execution

1. Open **Test your agent**.
2. Start a new conversation.
3. Ask: **Show my employee information**
4. Enter a valid Employee ID.

Example: **EMP0001**

The screenshot displays the 'Enterprise HR Assistant' interface. On the left, the 'Test your agent' panel shows a workflow diagram: 'Employee Lookup' (Topic, 7.71s) leads to 'Get a row' (Connector Action, 2.35s), which is marked as 'Completed'. On the right, the chat interface shows a conversation where the user asked 'Show my employee information' and provided the ID 'EMP0001'. The assistant responded with the employee details for EMP0001.

Employee Profile	
Field	Details
Employee ID	EMP0001
Name	Meera Kulkarni
Email	meera.kulkarni1@contoso.com
Department	Operations

5. Verify that the employee details are displayed.

Explanation

This validates the Excel connector integration.

Step 29: Test Multiple Employees

Repeat the lookup using different Employee IDs.

Examples:

EMP0002

EMP0004

Explanation

Testing multiple records confirms that the lookup is dynamic rather than hardcoded.

Step 30: Test Invalid Employee ID

Enter an ID that does not exist.

Example:

EMP9999

Enterprise HR Assistant

OverviewKnowledgeToolsAgentsTopicsActivityEvaluationAnalyticsChannels

QPublishSettings...Test

Test your agent

Employee Lookup

7.71s

Get a row

Connector Action

2.35s

Get a row

Connector Action

2.36s

Get a row

Connector Action

1.97s

Get a row

Connector Action

Failed

Email

anita.joshi3@contoso.com

Department

Security

Location

Delhi

Manager

Manager 34

Employment Status

Probation

Leave Balance

12 Days

Is there anything else you'd like to know or any other assistance I can provide?

Just now

EMP9999

Error Message: The connector 'Excel Online (Business)' returned an HTTP error with code 404. Inner Error: No row was found with id 'EMP9999'. Error Code: ConnectorRequestFailure Conversation id: 3ad7164a-bda6-49c8-abdc-e341d250f0dc Time (UTC): 2026-07-05T10:27:58.403Z

Just now

Ask a question or describe what you need

0/2000

You're testing your agent's real responses and capabilities. Find troubleshooting help here. Make sure AI-generated content is accurate and appropriate before using. See terms

Verify that the agent handles the missing record gracefully.

Note: If no custom error handling has been implemented, the connector may return its default error message. This is acceptable for this introductory lab.
